2025-2026-1 Complex Analysis Final (Recalled Version)

1. Let u be a harmonic function, and f be a holomorphic function. Using $\Delta = 4\partial\bar{\partial}$, prove that

$$\Delta(u \circ f) = ((\Delta u) \circ f) \cdot |f'|^2.$$

2. Use the contour integration to compute

$$\int_0^\infty \frac{1}{x^a(1+x)} \mathrm{d}x, \quad -1 < a < 1.$$

3. 1) Let f be a holomorphic function on the unit disk $\mathbb{D}$. Suppose that $|f(z)| \leq M$ on $\mathbb{D}$ and $f(a_j) = 0$ for some $a_j \in \mathbb{D}, 1 \leq j \leq n$. Show that

$$|f(z)| \leq M|\frac{z-a_1}{1-\bar{a_1}z}| \cdots |\frac{z-a_n}{1-\bar{a_n}z}|, \quad z \in \mathbb{D}.$$

   Hint. Use the same method for proving the *Schwarz Lemma*.

   2) A *finite Blaschke product* is a rational function of the form

$$B(z) = e^{i\varphi}(\frac{z-a_1}{1-\bar{a_1}z}) \cdots (\frac{z-a_n}{1-\bar{a_n}z}),$$

   where $a_1, \cdots a_n \in \mathbb{D}$ and $0 \leq \varphi < 2\pi$. Suppose that f is holomorphic on $\mathbb{D}$ and continuous on $\bar{\mathbb{D}}$ with $|f(z)| = 1$ for $|z| = 1$. Show that f is a *finite Blaschke product*.

4. Let f be a holomorphic function in the punctured unit disk $\mathbb{D}^*$. Suppose that there is a sequence of poles $\{z_j\}_{j=1}^\infty$ of f which accumulates to the origin. Prove that $f(\mathbb{D}^*)$ is dense in the complex plane $\mathbb{C}$.

   Hint. Use the same method for proving the *Casaroti-Weierstrass Theorem*.

5. Suppose Ω is an area in $\mathbb{C}$, and 0 is on the boundary of Ω. Suppose the interval $[0, 1]$ is in $\mathbb{C} \setminus \Omega$, construct a subharmonic barrier near 0.